Tesfatsyon Simon **Ochono**

PhD Candidate | Civil and Environmental Engineering (Smart Cities) | Western University

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ORCID | Google Scholar | LinkedIn | ResearchGate

PROFILE

Researcher in building energy and climate adaptation, developing design methods for energy-efficient, thermally comfortable and low-carbon buildings and data centers. Integrates building information modeling, building performance simulation, multi-objective optimization and machine learning to support robust decisions under present and future climates.

RESEARCH INTERESTS

- Building performance simulation
- Building thermal performance
- Heat transfer and thermo-fluids
- Climate-adaptive design
- Design optimization
- Machine learning
- Building digital twins
- Energy-efficient data centers
- Low-carbon buildings

EDUCATION

PhD, Civil and Environmental Engineering (Smart Cities) 2023 – Present
Western University *London, ON, Canada*

- Thesis: Integrated Design Optimization for Sustainable Buildings and Data Centers Using Physics-Based Simulation and Machine Learning Under Climate Change
- Supervisors: Prof. Girma Bitsuamlak and Prof. Katarina Grolinger
- CGPA: 90%

MSc, Power Engineering (Thermal) 2020 – 2022
Warsaw University of Technology, Faculty of Power and Aeronautical Engineering *Warsaw, Poland*

- Thesis: Extracting Heat from Nuclear Power Plants for District Heating: A Case Study of the AP1000
- Supervisor: Prof. Rafał Laskowski
- CGPA: 4.62/5.00 (92.4%)

BSc, Mechanical Engineering 2012 – 2017
Arba Minch University *Arba Minch, Ethiopia*

- CGPA: 3.61/4.00 (90.25%)

TEACHING EXPERIENCE

Graduate Teaching Assistant Jan 2024 – Present
Western University *London, ON, Canada*

- Led tutorials and laboratory sessions; graded assignments, reports, quizzes and examinations
- Held office hours; invigilated quizzes, midterms and final examinations
- Courses:
  - Heating, Ventilating and Air Conditioning
  - Computational Tools for Civil Engineers
  - Climate Hazards for Engineers
  - Building Sustainability
  - Foundations of Engineering Practice

Assistant Lecturer Dec 2018 – Jun 2019
Department of Mechanical Engineering, Wolaita Sodo University *Wolaita Sodo, Ethiopia*

- Assisted in teaching Engineering Thermodynamics and Heat Transfer
- Advised students; supported departmental research, workshop practice and community service

PROFESSIONAL EXPERIENCE

Senior Process Executive (Programmer) Jan 2022 – Sep 2023
Cognizant Technology Solutions (client: Google) *Kraków, Poland*

- Evaluated cases for the Google project against detailed policy guidelines
- Identified complex issues and referred them for senior review
- Achieved high accuracy and quality while meeting service-level targets

PUBLICATIONS

[C2] **T. S. Ochono**, A. Awol, K. Grolinger and G. T. Bitsuamlak. Energy optimization approach for mass timber office building design under climate change. In *Proceedings of eSim 2026: 14th Conference of IBPSA-Canada*, pp. 93–97. IBPSA-Canada. doi:10.26868/31114504.2026.232 — 2026

[J1] J. M. Hunde, **T. S. Ochono**, D. Senevirathne, D. D. Eneyew, G. T. Bitsuamlak, M. A. M. Capretz and K. Grolinger. Data-driven and physics-based modeling approaches and their integration in building digital twins: A systematic review. *Journal of Building Engineering*, 114, 114214. doi:10.1016/j.jobe.2025.114214 — 2025

[C1] J. M. Hunde, P. Manmatharasan, D. Senevirathne, **T. S. Ochono**, D. D. Eneyew, G. T. Bitsuamlak, M. A. M. Capretz and K. Grolinger. Predicting building energy demand using federated learning with attribute-based clustering. In *Proceedings of the 9th International Building Physics Conference (IBPC 2024)*, Lecture Notes in Civil Engineering, pp. 50–56. Springer. doi:10.1007/978-981-97-8309-0_7 — 2024

CONFERENCE PRESENTATIONS

Energy Optimization Approach for Mass Timber Office Building Design under Climate Change — 2026
eSim 2026: Advancing Sustainability and Climate Resilience, Longueuil, QC, Canada

Predicting Building Energy Demand Using Federated Learning with Attribute-Based Clustering — 2024
9th International Building Physics Conference (IBPC 2024), Toronto, ON, Canada

Application of the AP1000 Nuclear Power Unit to Supply Heat to the District Heating System — 2023
XV Youth in Power Engineering Conference, Wrocław, Poland 3rd Place, Best Presentation Award

AWARDS AND HONORS

3rd Place, Best Presentation Award, XV Youth in Power Engineering Conference, Wrocław, Poland — 2023

SPINAKER Scholarship, Polish National Agency for Academic Exchange (NAWA) — 2022

Ignacy Łukasiewicz Scholarship, Polish National Agency for Academic Exchange (NAWA) — 2019

SKILLS

Simulation software	DesignBuilder, EnergyPlus, OpenStudio, IES VE
CFD	STAR-CCM+, OpenFOAM, Ansys Fluent
Programming	Python, C/C++, MATLAB
Machine learning	Surrogate modeling, time-series forecasting, federated learning
Optimization	Multi-objective and constrained design optimization
CAD and BIM	Autodesk Revit, AutoCAD, SketchUp, Onshape
Research computing	High-performance computing (SHARCNET, Compute Ontario)
Languages	Amharic (native), English (professional, C1), Polish (intermediate, B1)

CERTIFICATIONS AND TRAINING

DesignBuilder Simulation Training, DesignBuilder Software (48 CPD hours) — 2025
Building energy modeling, HVAC systems, loads, daylighting, CFD and optimization

Compute Ontario Summer School, ACENET, SciNet and SHARCNET — 2025
Introduction to Advanced Research Computing; Working with Jupyter on the Clusters; Scaling Up HPC Workflows

Parallel Programming with C++, SHARCNET — 2023

SERVICE AND MEMBERSHIPS

Volunteer, 8th International Symposium on Computational Wind Engineering (CWE 2026) — 2026

Member, International Building Performance Simulation Association (IBPSA) — 2026 – Present

Member, American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) — 2026 – Present

References available upon request.